

## RESEARCH ARTICLE

# Shaping futures: A dialogue on chemical engineering education

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## Abstract

Engineering as a discipline, profession, practice, and area of study continues to add substantial value in an increasingly complex world. With continually evolving complexity around the planet, such as the need for massive energy transition, global health technologies, or sustainable food systems, how might engineering education practices and theory be considered within these rapid and necessary changes? This paper presents an experiment of co-creation through experiential reflection about the state of chemical engineering education. Four chemical engineering professors engaged in a dialogue, facilitated by a researcher in education, through collaborative and actionable research. This dialogue uncovered innovative possibilities, educational themes, experiences, and opportunities for others in the profession to consider. The process of dialogue also encouraged the development of an imaginative future sense-making, known as futuring, through a collective experience. The findings reveal instructive perspectives on the shape of chemical engineering education that should be of value not only to engineers, but also other professionals, practitioners, or those in various science, technology, and math fields.

## KEYWORDS

chemical engineering education, collaborative research, complexity education, dialogue, futuring, reflective practice, systems approach

## 1 | INTRODUCTION

Is chemical engineering still relevant in the 21st century? While society's opportunities and challenges have shifted, chemical engineering does indeed remain one of the vital disciplines of our time, and is critical in shaping societal

futures. It is not only central to technological advancement in key areas, such as energy, medicine, and sustainability,<sup>[1]</sup> but also has the capacity and potential of educating students for adaptive changes amidst significant societal transitions. With continually evolving complexity around the planet, the primary question addressed in this paper is how might

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we reconsider chemical engineering education practices and theory within these rapid changes?

## 1.1 | Context

Given the complexity of global issues, and the exponential advancement in technology and informatics, engineering curricula are seeing enormous challenges in evolving to stay relevant to equip graduates.<sup>[2,3]</sup> Some educators are responding to this transition by rethinking what and how to teach, but actual change is happening too slowly. Visionary engineering educator Kamp<sup>[4]</sup> observed: 'It is so much easier to educate students for our past than for their future' (p. 3). Curriculum assessment and reform is always a difficult change process, even if it only involves making minor changes, as most engineering curricula are already overloaded. The challenge is amplified when major change is long overdue. Armstrong noted how little chemical engineering curricula have changed over four decades (1965–2005), particularly after its initial rapid evolution from 1905 to 1965, despite substantial changes across industry and the discipline in general.<sup>[5]</sup>

More recently, Voronov et al. compared core chemical engineering courses in 148 curricula across the United States and found how remarkably similar the courses are to curricula from 50 years ago.<sup>[6]</sup> Agrawal et al.<sup>[7]</sup> examined curricula of six chemical engineering programs from three Washington Accord countries—England, South Africa, and the United States—focusing on contact hours, curricular rigidity, and first year structure. Based on the differences in curricula and student experiences, they raise questions as to whether high contact times and curricular rigidity are truly necessary.

This is not to say that no changes have occurred.<sup>[8]</sup> Over the last decade there has been increased focus on engineering education research and the scholarship of teaching and learning that has supported valuable evolution at most Canadian universities in the instruction of communication,<sup>[9]</sup> leadership,<sup>[10–12]</sup> problem solving,<sup>[13]</sup> design,<sup>[14]</sup> and lab-based-instruction,<sup>[15,16]</sup> among others. However, most of these initiatives have been attempts to fit more into already overfilled curricula, rather than undertake major curriculum reform.

Some universities are considering more fundamental and substantive change, prompting valuable discussion of the desired skills and attributes of a 21st century chemical engineer. Fletcher et al. surveyed 94 newly employed and upper year chemical engineering students at the University of Strathclyde regarding the importance of 24 skills to their work.<sup>[17]</sup> Transferable skills were

perceived as most important, matching the findings from a larger 2004 survey of 2158 participants from 63 countries by the World Chemical Engineering Council. Here, respondents also reported that their capacity to apply knowledge and basic chemical engineering fundamentals exceeded their importance in the workplace. In contrast, they reported learning deficits for many transferable skills that they perceived to be highly important for employment success such as the ability to solve problems, work effectively in a team, and learn independently.

Similarly, complementary/professional skills are increasingly valued by Canadian employers.<sup>[18,19]</sup> This is not to say that technical skills are unimportant. Rather, graduates are less competent in complementary skills and that these are stronger determinants of career success. Teaching technical content is highly emphasized in most Canadian engineering programs. For example, in a review of a Biosystems program, Seniuk Cicek found a significant disparity between the emphasis placed on the Knowledge Base for Engineering, constituting nearly half of the program, in contrast to the 9% importance attributed by stakeholders.<sup>[20]</sup> Conversely, individual and teamwork, identified as the most crucial attribute by stakeholders at approximately 11%, received just over 1% attention in the program.

With the desire for greater consistency in learning outcomes nationally and internationally, other groups around the globe are also looking to modernize chemical engineering curricula, such as in Europe,<sup>[21,22]</sup> Brazil,<sup>[23,24]</sup> Columbia,<sup>[25]</sup> Uruguay,<sup>[26]</sup> China,<sup>[27]</sup> and South Africa.<sup>[28,29]</sup> Common themes include the diminishing relevance of traditional curricula, the lack of sufficient training in transdisciplinary competencies and transferable/success skills such as communication, collaboration, leadership, and lifelong learning, and the need to better prepare graduates to tackle complex, ill-defined problems.

More broadly, other research has outlined the need for transdisciplinary forms of engineering education, for instance, as a way of educating for complexity.<sup>[30]</sup> Martin et al. asserted how engineering education must be adaptive in this environment of uncertainty and rapid change.<sup>[31]</sup> In particular, they focus on the need to link engineering design with climate and sustainability contexts, incorporate various disciplines into engineering solutions, consider justice and ethics dimensions, and collaborate with diverse communities. Other approaches, such as engineering leadership,<sup>[11,32,33]</sup> engineering design,<sup>[34,35]</sup> and emerging learning environments,<sup>[36]</sup> offer integrated forms of education.

This paper explores these and other considerations related to chemical engineering education from a process-based research method of a collective dialogue.

Educational research within the field of engineering is experiencing substantial growth, and understandably so—nurturing the skills and knowledge of the next generation of engineers is paramount, and these include integrative, transdisciplinary, systems-based, and interactive forms of education. However, many educators are seeking guidance in navigating these intricate practices. Within the constraints of curriculum governed by engineering accreditation and high number of hours that are accounted for, simply reimagining chemical engineering curriculum seems insurmountable. Rather than presenting a programmatic paper, the intention is to embark on a distinct form of inquiry that can model a type of investigation into chemical engineering education, serving as an open inquiry into how shaping the futures may unfold.

## 1.2 | Outline of approach

The concept of dialogue, defined as the act of suspending assumptions to collaboratively generate new meanings through communication,<sup>[37]</sup> was used as the primary methodology underlying this paper. Dialogue was approached as an action-oriented methodological framework that fosters the creation of meaning through a collective and shared experience. The focus involved a facilitated dialogue by a researcher in education with four colleagues and collaborators who are experts in the field of chemical engineering. This exchange was organized around three overarching research questions:

1. How has chemical engineering education evolved as an instructional practice?
2. How can chemical engineering education adapt to the increasing complexities of our world?
3. In what ways might chemical engineering education equip students for the challenges of future worlds?

These questions are presented as invitations to encourage participants—that is, the authors of this paper and, by extension, other readers—to engage in dialogue as an exploratory approach to action research. The intent is to stimulate reflective conversations and collectively contribute to the ongoing evolution of chemical engineering education.

Through collaborative discourse among experts in the field of chemical engineering and their broader contemplation of engineering education, the approach is to cultivate an environment that challenges conventional boundaries within scholarship. Rather than engaging solely in theoretical discussions or practical applications, the aim is to actively engage in a nuanced dialogue, inspired by Meadows' assertion that actionable movement

is essential to change.<sup>[38]</sup> This approach will facilitate a comprehensive understanding of the potential trajectories that the discipline of chemical engineering might undertake.

The approach is supported by incorporating the educational concept of 'futuring.' This term denotes a continuous, collaborative process where participants envision, imagine, and shape potential futures. Futuring offers an open-ended way of sense-making around a subject, experience, and/or interaction, thereby understanding and actively contributing to various potential futures.<sup>[39]</sup> Through engaging in dialogue that recognizes temporal interconnectedness, these educational research efforts aim to foster opportunities for future exploration and advancements in the field without making prescriptive proclamations.

In the following pages, chemical engineers with extensive expertise engage in a dialogue through collaboration and actionable research, to uncover and explore possibilities for others in the profession to consider in terms of educational themes, experiences, and opportunities. The paper includes a theoretical section that considers the notion of dialogue as an educational practice of un/learning, followed by the methodology and methods of action research supported by collaborative and autoethnographic inquiry. The results of convergence of this paper emerged out of both an in-person and written dialogue by the four chemical engineers facilitated by an education researcher. It concludes with a reflection section that provides suggestions for future considerations that emerged from the dialogue.

## 1.3 | Theoretical framework

Communication and dialogue not only present ways of exchanging ideas, but also co-create meaning and knowledge. When speaking to wide ranging audiences—as the authors attempt to do here in chemical engineering, other engineering fields, and in sciences, technology, engineering, and mathematics (STEM) education more broadly—creating spaces to explore in process-based and qualitative research is becoming increasingly important. By 'process-based', this paper refers to cultivating space and time in order to explore and generate learning without preconceived targeted outcomes. Structured rather than more open-ended knowledge production can often hinder creative potentials for creative future building.

This use of the word 'process' is admittedly different from the usage that most chemical engineers are familiar with, where a desired product is known and the process of optimizing is typically the goal. However, some analogies exist. For example, when students go out on co-op positions, the focus of the learning is open ended

and is achieved through the process, rather than by working under predetermined conditions or producing a set product. A more distant metaphor might be made in terms of Reynolds number and turbulence. By using dialogue, the aim is to increase the 'driving force' for the flow of ideas to overcome the static resistance—or viscosity—created by our own preconceptions and unrecognized understandings. The result may at times have been more chaotic, but turbulence and greater mixing is not necessarily a bad thing.

This conceptual approach is inspired by the theoretical framework articulated by the physicist and theorist Bohm,<sup>[37]</sup> centring on the creation of knowledge through dialogue. This framework proposes a process-oriented approach to conducting educational research, emphasizing discursive explorations. Unlike common parlance of the term discussion—which shares the same etymological root as the words percussion and concussion, meaning to break up or analyze—dialogue derives from the Greek word *dialogos*. Breaking it down, *logos* means 'the word' but refers more broadly to 'meaning of the word', whereas *dia* means 'through', rather than the misconstrued perception of 'two'. Thus, *dialogue*, according to Bohm's analysis, is defined as the 'stream of meaning flowing among and through us and between us' (p. 7).<sup>[37]</sup> When engaging in dialogue, as opposed to discussion, flow of meaning and new understandings arise *through* the collaborative potential of the group or collective.

Within educational settings, the concept of dialogue, also known as dialogic practice or open dialogue,<sup>[40]</sup> emerges as a valuable framework. It extends an invitation for active participation in the collaborative construction of meaning through language. Moreover, it provides a process-oriented, co-creative approach that underscores the relational and meaning-making aspects of language and communication. As cautioned by Wittgenstein,<sup>[41]</sup> meaning does not simply emanate from an individual's private understanding; rather, it emerges from coordinated action through a shared understanding of language and communication, and these collective relationships shape our worlds.<sup>[42]</sup> Using dialogue is another possible way to increase the 'driving force' for the flow of ideas to overcome the static resistance created by our own preconceptions and private understanding.

Participation in such dialogue requires the capacity to suspend assumptions, enabling the emergence of a collective response. According to Bohm,<sup>[37]</sup> individuals, each bringing diverse backgrounds, harbour varying degrees of assumptions and opinions. In contemporary society, a significant challenge lies in the ability (or lack thereof) to coexist with differences. The pivotal question becomes: How can we create a space that accommodates multiple perspectives? This fundamental aspect of

dialogue forms the bedrock of generative pedagogy, emphasizing the importance of fostering an environment that encourages the exploration and appreciation of diverse viewpoints.

Dialogue is for connection and in contrast to individual arguments protecting a sense of knowledge or learning that has been individualized. Co-creation builds knowledge together, specifically on dialogue, that this is the very nature of dialogic practice. Educators can craft or sculpt dialogue in practice, in various spaces, and in academic and/or educational spaces, and such dialogue can invite various forms of inquiry. Thus, this paper demonstrates dialogue as a way of learning from, with, and through each other, as well as modelling this for others as an invitation to build new or reinvented forms of knowledge around engineering education. To enter in a dialogue is to also co-create new forms of knowledge that are adaptable and responsive to applied practices. While offering expansive ways of examining knowledge and experience, dialogue can also be non-linear, emergent, and uncertain—that is, it can be complex.

In this sense, encouraging dialogue as a form of educational inquiry corresponds well with complexity education, which considers dynamics in education that inquire into non-linear and emergent properties of meaning.<sup>[43–45]</sup> As the dialogue demonstrates, whether the topics are about artificial intelligence (AI), technology, collaboration, learning and assessment, or diversity, equity, and inclusion, ever increasing complex systems intersect. While the entropy of the universe is always increasing, returning to the concept of convergence, complex systems can still come together.

Building on Morrison,<sup>[46]</sup> the following questions emerge as background to the dialogue in the next section: What does it mean to know or be in a time of uncertainty? What creates a web of learning with rich mental maps rather than linear, programmed learning? What conditions might cultivate an education that is complex and emergent? How might post-secondary programs be developed as well-integrated yet adaptive systems, which move outside of measuring performance, assessment, and predictable pedagogies? To what extent should our goal as educators be to share knowledge versus help students develop for success?

## 2 | METHODOLOGY

The intent of this paper is to explore and co-create meaning around our experiences through engineering education and guide the futuring process. Thus, a qualitative research approach was applied to surface and develop an in-depth understanding of current engineering education.



Inquiry-based research was chosen as the research approach, particularly collaborative inquiry that draws on autoethnographic experiences and interactions of the participants in the research. This allowed the authors to observe and respond to developing ideas, concepts, and pedagogies, while being actively engaged in shifts and changes through collaboration. Knowledge, in collaborative inquiry, is considered to be created through the co-action of the participants/researchers. Based on this, the notion of dialogic inquiry was used as a method for joint construction of meaning to emerge.

Rooted in the action research framework of ‘transformative social learning with a change agenda’ (p. 7),<sup>[47]</sup> we surfaced understandings, reflections, and explorations of our professional experiences as engineering educators in practice. Using methods of participant observation, group discussion, and personal and collective reflections, we weaved meanings among the threads of experience.<sup>[48]</sup> Through critical reflections based on inquiries, we aimed to transform our ways of knowing<sup>[49]</sup> about the current state of engineering education, while demonstrating these shifts as the ‘results’ of the dialogue in this paper. Action research ultimately allowed the possibility for social change to be produced in process or motion. The three inquiry questions that frame this dialogue arose from the epistemologically informed and interdisciplinary backgrounds of the four participants in chemical engineering education.

## 2.1 | Participants

The participants included four professors teaching chemical engineering in Canada. One professor in education facilitated the dialogue and framed the theoretical and methodological process. All authors have backgrounds in sustainability and transdisciplinary education, with expertise in cross-disciplinary collaboration.

## 2.2 | Data collection

Our collaborative inquiry followed Larrivee’s critical reflection framework.<sup>[50]</sup> First, we examined our current practices through personal reflections and existing literature in a video-conference call together. Second, we collected and collated responses from the video call of the four engineering educators. Third, experiences, reflections, thoughts, and literature based content were added through a process of collaborative writing, where distinct voices were blended as a collective response through three iterations of writing and revision. Fourth, we had another video call to initiate the collective visioning

process through discussing key topics. Fifth, all the authors engaged in documenting the collective visioning of how to transform our practices and future engineering education in the Dialogue section. The output of this process is documented as the dialogue with further considerations outlined in the closing Reflections section.

## 3 | THE DIALOGUE

### 3.1 | How has chemical engineering education evolved as an instructional practice?

Chemical engineering education has evolved over the last decade in terms of the introduction of design courses across the curricula of many undergraduate programs. This often includes instruction of general design principles in the first year of an undergraduate education, often with more chemical engineering specific design in middle years, leading to a longstanding capstone course in the final year. Many or most chemical engineering students now participate in co-op and internships programs (Liu et al., 2018).<sup>[51]</sup> Instruction of transdisciplinary competencies, such as communication and more recently team-skills, for example,<sup>[52]</sup> has also been added to many curricula.<sup>[53]</sup> Active-learning-based instruction is growing but remains the exception more than the rule across courses and instructors.

The traditional undergraduate curriculum has overwhelming emphasis on transmission of technical information in most classes. For example, the curricular emphasis on traditional fundamentals of chemical engineering, such as transport theory, separation processes, and chemical reactor design, has not evolved much over the last 50 years. While these fundamentals help students see the world as a chemical engineer, the heavy focus on these fundamentals represents an outdated understanding of what many or most students do after graduation. Holding on to this traditional identity is creating a quandary as the discipline is struggling to shift from the past and pave the way for a different future. And yet, evolving and redefining is necessary to create the space and flexibility needed for chemical engineering. This barrier was noted almost two decades ago by Armstrong,<sup>[5]</sup> among others, and is indicative of the challenges to actually implement substantive forward looking curricular change.

Over the past few decades, the outcomes assessment criteria (graduate attributes and continual improvement) by Engineers Canada have supported the way we structure and scaffold courses. As a result, engineering education has become more intentional and explicit

around instructional constructive alignment. The 12 graduate attributes (GA), including design, team work, communication, impact of engineering on society and the environment, and ethics and equity, have guided and allowed curriculum to shape instructional practice. Topics such as ethics/professionalism and impacts of technology on society used to be taught in isolated and specialized courses. Now there are efforts being made to integrate these topics, as well as those related to equity, diversity, inclusivity, and Indigeneity, into more courses.

Problems that engineers need to solve are increasingly requiring holistic and integrated ways of understanding. This has required us to change the instructional practice to bring about context and impact of engineering solutions mainly through design. For example, techno-economic analysis (TEA) has been a standard tool for comparative evaluation and making process design decisions for decades. More recently, the TEA has taken on new requirements with respect to triple bottom line analyses (i.e., social, environmental, and economic). Sustainability and net social benefit analysis and eco-TEA incorporating life cycle analysis are becoming the in-demand skill set as engineers need to negotiate the complex socio-enviro-technical and systemic aspects of engineering work.

More faculty members are considering how course curricula and activities are being designed and seeking to better understand student experience through surveys/data collection. The introduction of the teaching faculty stream (e.g., Educational Leadership at the University of British Columbia, Teaching Professor at University of Alberta, Professor Teaching-Stream at the University of Toronto) is reflective of the commitment and investment in education being made at many Canadian universities, although this change may also in part be a way to mitigate the impacts of the increased enrolment in engineering programs needed to meet budgetary pressures. Even more traditional research stream faculty have embraced engineering education as a beneficial course to improve program development and student recruitment.

Faculty are more aware of the opportunities for improvement, such as incorporating more active learning in lecture-style classes, understanding how group work can be better structured to improve individual learning in project/lab-based courses, and integrating more inquiry- and discovery-based learning into courses. Funding is also being made available to enable and incentivize these opportunities. This support for instructional training and educational research funds is already paying dividends through the quality of courses and innovative pedagogical approaches taking place.

Furthermore, the strategic priorities of our Departments, Faculties, and Universities have also evolved to emphasize the community and global awareness of our graduates and their aptitude to take on the increasingly complex challenges facing society. This then creates the opportunities for our instructional practices to pursue new curricular and co-curricular activities for our students to grow in these areas. There is a lot of exciting work to be done and interest and support to do it.

Students are underprepared for the integrative thinking and synthesis required for their senior design courses. Since they study fundamentals in siloed courses, students can be overwhelmed when equipment and process design requires them to remember and integrate the fundamental concepts in a new way. Earlier integrative design experiences would reduce this cognitive burden on students. As most faculty do not teach design, they may not appreciate the challenge students face and it is not something one can learn by reading a textbook. Synthesizing declarative and procedural knowledge acquired through fundamental courses requires skills that are often developed experientially and through practice.

Another evolution of instructional practice resides in the shifting internal and external dynamics affecting students. Coupled with many factors, including the pandemic (2020–2022), global conflicts, and climate change, we have seen anecdotally—that is, through our own experiences and in discussions with colleagues—how students have been experiencing increased challenges with mental health and in-class behaviour. Suggestions are that this is a global phenomenon and while the underlying causes are presumably multifold, Ucher has noted, based on declining scores on the Programme for International Student Assessment globally, that this change is correlated with the introduction of smartphones a decade ago.<sup>[54]</sup>

According to Lukianoff and Haidt,<sup>[55]</sup> there is also a trend in students demanding to remove words and ideas that are uncomfortable from campuses. The reasons for this trend are multifold, including generational shifts and societal changes concerning safety. The generation currently in university has been dubbed ‘Generation Z’ by some and ‘Generation C’ by others. Broo et al. documented<sup>[56]</sup>: ‘Its members have one big thing in common: they are digital natives and exceptionally tech-savvy, staying connected all the time. Additionally, they are Communicating, Content-centric, Computerized, Community-oriented, and Clicking’ (p. 3–4). These trends have not only shifted campus culture, but also required instructors to adapt teaching and pedagogy. An empowering change within this cultural shift is also how the students have an increasing awareness of and identification with

diversity, including more participation of female students and students from under-represented cultural backgrounds.

Students find themselves immersed in an information-rich environment, often inundated by social media and an unceasing news cycle. This perpetual exposure has cultivated a heightened awareness among students regarding the myriad of issues affecting society. A growing number of students express a profound sense of responsibility to actively engage in addressing these global challenges. However, this sense of duty also introduces an additional layer of pressure. Coupled with the inherent demands of an already rigorous curriculum, this heightened responsibility can contribute to an increased burden.

Nearly 20 years ago, the documentary *An Inconvenient Truth*<sup>[57]</sup> sparked an awareness among students, compelling them to consider the potential impact of climate change and their role in addressing it, particularly as they contemplated their future careers. In contrast to the cohort nearly two decades ago, students now find themselves unable to escape the omnipresence of climate change. Educators can gain new awareness by adapting to this new reality. Striking a balance between fostering social consciousness and ensuring the well-being of students becomes a pivotal aspect in navigating the intricacies of contemporary chemical engineering education.

### 3.2 | How can chemical engineering education adapt to an increasingly complex world?

Faced with myriad societal challenges, engineers would continue to benefit from cultivating the ability to deal with complexity—that is, dynamic, contextual, value-laden, uncertain, emergent, self-organizing properties as part of understanding the world—and shaping their worldview. The roles taken on by graduates, along with their career paths, are much more diverse than they were decades ago, with most graduates no longer working in traditional chemical engineering industries. In fact, Canada graduates 1300 chemical engineers a year for only 300 job opportunities in traditional chemical engineering industries.<sup>[58]</sup> Moreover, even if graduates are working in the traditional energy and chemical industries, there are new demands with respect to the energy transition, economic transition, data analysis, AI, and managing social and geopolitical complexity. Engineering education must adapt to help prepare graduates for success in these diverse careers and equip them to tackle these increasingly complex roles.

The common notion of ‘T-shaped professionals’ has entered the engineering education conversations to deal with detailed complexity (technical depth) and dynamic

complexity (evolving contextual nature) in which professionals are required to operate.<sup>[59]</sup> Many engineering educators talk about the need for ‘T-shaped engineers’ without understanding or knowing how engineering education can shift to produce them. In the context of sustainability, for example, siloed technical knowledge needs to be connected to a larger context beyond engineering. More generally, transdisciplinary instruction (e.g., structuring curricula around larger human-centred contexts that transcend disciplines such as sustainability) can help students better inter-connect and integrate knowledge so that it can be more readily transferred and applied in other contexts. In fact, we are starting to see a few examples of courses/programs that have students from different disciplines working together (e.g., Humanitarian Engineering course,<sup>[60]</sup> Transdisciplinary PhD program<sup>[61]</sup>) as well as more emphasis on systems-based approaches to complex problem analysis at various levels throughout our program.

Students and faculty may benefit from incorporating various ways of knowing (epistemology) and considering multiple worldviews (ontology), particularly by seeing a world where evidence and expertise are less valued or accepted as the basis for decision making. The spread of misinformation or ‘alternate facts’ will presumably continue to grow, potentially at an accelerating rate due to generative AI, which will further undermine the value placed on university education and the associated credentials. To address this, graduates will need to be better able to analyze and openly critique incorrect information and make compelling arguments for why their knowledge and experience should carry more weight. This may appear to create a contradiction with the need to incorporate more diverse ways of knowing but this is a false paradox, as types of knowledge are all built from foundations of shared evidence whether this is observational, empirical, or theoretical. Graduates will be expected to recognize, assess, and integrate many types of evidence and recognize which is best suited for a given context, so as to put forward recommendations and decisions based on the best available knowledge. They will also require effective communication skills to become more convincing and persuasive in facing misinformation or conspiracy theories. Again, this speaks to the need for graduates to learn both continuously and broadly—the T-shaped graduate.

Building on these expanding forms of education and training, engineering education should move towards more holistic learning experiences that emphasize the acquisition of essential skills and competencies, along with a more general technical knowledge base to allow students to dissect and frame problems, with less focus on acquiring deep technical knowledge that only a few

graduates will ever use. Although it is challenging and can be uncomfortable for students, confronting complex problems early in their engineering program is a necessity. It provides a framework for students to connect the technical skills they are learning to in the broader picture.

For example, most chemical engineering graduates may never design distillation columns or chemical reactors upon graduation. However, students still need to be able to construct a reaction network and think through whether the reactor is kinetically or thermodynamically controlled. They still need to use a process simulator and validate results. In other words, they still need to understand the underlying principles so that they can transform this knowledge to new situations. There is clearly a need for some procedural knowledge—that is, how do I do *x*—but too much of this can lead to rote learning. Students tend to crave procedural knowledge because they often equate being able to apply knowledge with understanding it (e.g., “I can calculate Gibbs Free Energy so therefore I understand the second law of thermodynamics”).

The rapid emergence of AI will further reduce the need for procedural knowledge, often acquired by students through rote learning, and increase the need for conditional knowledge that will help students tackle new applications. An appreciation of the value of deep knowledge and the limits of knowledge and expertise will still be important, but AI will create a greater need for graduates to be able to navigate uncertainty by asking the right questions to prompt the AI rather than knowing answers. Critical and creative thinking accompanied by collaborative learning and effective communication will be the necessary skill set for graduates.

Promoting critical and creative thinking is also central to training engineering students for a complex world, rather than training them to use predetermined algorithms for problem-solving or blindly follow design heuristics. These can provide a starting point, but students need to be comfortable adapting these for a given context, and that means being able to manage new and uncertain situations. This has always been the case in engineering, but with limited time and a lot of content to cover across the degree program, we have had to ensure we have provided students with the information they need to solve problems. Transmitting information and solving example problems was seen as the efficient way of doing this.

With the emergence of AI technologies, it seems that problem-solving methodologies, heuristics, and design equations will be available at their fingertips. Such accessibility may allow us as educators to free up time for deeper learning that can emerge from grappling with new and unfamiliar problems. Rather than spending extended periods of time teaching a multi-step procedure for designing a piece of equipment,

we can spend more time thinking about how that procedure may need to be adapted for different or more complex situations.

Another essential dimension of this evolution will be helping students ‘learn how to learn’ by equipping them with the metacognitive skills they will require to learn efficiently and effectively, and transfer knowledge to new situations. As society evolves from a knowledge-based to a learning-based paradigm, individuals and organizations that can learn quicker will be more successful. AI will accelerate the capacity of anyone to generate existing knowledge building on established connections, switching the role of engineers to creating new knowledge by synthesis, integration, evaluation, and validation, resulting in a continuous learning process. As students become more exposed to open-ended questions and start to understand there are multiple approaches thus solutions to a question, they can cultivate metacognitive skills to understand their learning. Pushing further, we have an opportunity to build a curriculum to ‘learn to co-learn’ as learners. The curriculum and programs can evolve to allow co-learning to occur, while the content is structured such that learners see how to connect the dots and avoid siloed concepts.

Taking a moment here to connect the dots of this dialogue in addressing how we train students for working in a complex world, it is interesting to note how systems approaches are not yet a normative part of engineering education. While considering holistic ways of thinking and learning, technical knowledge can be situated contextually in society, economy, community, environment, and other systems. At the program level, this may be a way to connect content and reduce the ‘hours’ required by accreditation bodies of engineering programs. Once again, this may look like challenging students with complex design problems (e.g., problem-based learning) that necessitates students understanding the contextual aspects of technology. Emphasizing the interconnectedness of various elements within a system provides an understanding of how technical interventions interface with and are implemented within a broader context that dictates their effectiveness. This can help students be better prepared to navigate the intricate landscape of real-world problem-solving, where solutions require a nuanced understanding of the interplay between engineered technology and its broader environment.

Introducing the concept of multiplicity and diversity in approaches and thinking will be important—knowing that there are many good solutions and there may be better solutions and not just one good solution. Acknowledging that many students are attracted to engineering because of the perceivable comfort in binary right/wrong answers (i.e., duality thinking), it may be necessary to



intentionally unlearn this notion, or even further explore relativism earlier on in their education. How do we hold space for multiple truths? This seems like a big challenge for engineering students. How does intellectual maturity play a role in their development in multiplicity?

We wonder if by emphasizing questions with single answer solutions, we are delaying this intellectual development. Another consideration is to integrate experiential and multidisciplinary learning to the program where students are faced with diverse ways of knowing, thinking, and perceiving. Another offshoot of these points raised in this section about holistic education and systems training begs the question: how do we equip students to deal with ambiguity and uncertainty that comes from complexity? This provides a segue to the final dialogue question.

### 3.3 | In what ways might chemical engineering education equip students for the challenges of future worlds?

This question really depends on how we envision future worlds, which brings in the concept of futuring that frames this dialogue. It is safe to say we are a more interconnected global community than ever before, which the pandemic highlighted, and that the challenges faced on one-side of the world affect and are affected by the other sides. Climate change does and will continue to affect various aspects of our lives, and in ways we may not yet recognize. This requires engineers to not only navigate uncertainty, but be equipped to produce adaptive and resilient solutions to forthcoming challenges. Acquiring technical knowledge should remain central to undergraduate engineering education but with the recognition that this knowledge base will primarily be used to facilitate and accelerate future learning over a lifetime, rather than being an end in and of itself that provides a static foundation that will be adequate throughout a career.

Engineering education can better prepare students for the challenges of the 21st century by increasing the emphasis on acquiring the skills and competencies already raised in the first two questions. From an education perspective, it means continuing to do the things that we are starting to see in our programs and curricula, which is integrating this learning throughout our courses rather than isolating them in specialized courses or technical electives. This also means getting our students to live the experience of problem solvers and not just grade getters, and it is our role to facilitate that. This means re-thinking the activities and tasks that we give our students so that in doing their assigned task, they have experiences that lead to the development of

skills needed for a more complex and interconnected world that awaits them. We are starting to see this through more transdisciplinary courses and programs,<sup>[62]</sup> and more experiential learning opportunities through project-based and community-engaged activities.

One of the challenges for educators is that these more integrative courses involving transdisciplinary perspectives and/or multiple stakeholders can be very difficult to implement for large class sizes, and difficult to evaluate when there might not be clear right/wrong outcomes for assessment. We need to determine if students have learned and grown as individuals through the activities we have assigned them. Reflection, through discussion or written assignments, is one way we can gather the meaning students are extracting from their experiences, and rubrics exist for evaluating depth of critical reflection. Integrating reflection assignments with relevant prompting questions, among other technical activities in our courses, may be a way to assess our students' development in these new growth areas that will be needed for working in a more complex and interconnected world.

The rapid development of AI-based natural language processing should open the door to assessing reflections and other qualitative text at the scale of a large engineering course. However, even with these new tools we will still need to push ourselves as educators to seek greater clarity of the learning and growth we are seeking to achieve and how we can recognize it. This is the fundamental challenge as it will require that we move beyond using solving a single answer problem as evidence of a student having learnt the related knowledge. While problem solving is a key attribute for engineering accreditation, these must be integrated with a more reflective worldview to understand the consequences to public good. El-Zein and Hedemann asserted the limitation of current focus on problem solving ethos in engineering education in dealing with complex societal challenges.<sup>[63]</sup>

Similarly, engineering programs should move towards more authentic assessment that places less emphasis on content acquisition and rote learning. This assessment could complement while reducing the emphasis on time-limited in person exams and tests, to better assess the highest-level cognitive skills and more closely represent the work that students will do and deliverables they will produce after graduation. Moreover, increased emphasis should be placed on assessment to provide formative feedback rather than assigning summative grades that have little meaning outside of academia. While in-person exams can still play a role to screen out students, there is no need for students to write 30–40 exams and an equivalent number of tests over the duration of an undergraduate degree in order to identify the 10% of students who are not suited for careers in

engineering. The value of using grades to motivate learning and rank students should also be reconsidered, as this often promotes short-term superficial learning (i.e., cramming) rather than long-term deeper understanding.

Another potential opportunity for engineering educators that has been hinted at throughout this dialogue is engaging more with multiple ontologies, epistemologies, and diverse cultural experiences and worldviews. Admittedly, these are areas often far outside of an engineer's training and expertise. Indigenization and decolonizing engineering education programs in Canada, even in minor ways, would have beneficial implications for future relationships with Indigenous groups and communities across Canada.<sup>[64]</sup> Whether these are linked through work in industry, government, or academia, building deeper relationships and understanding and listening to the needs of Indigenous communities remain crucial in decolonizing engineering practices.<sup>[65]</sup> This is particularly true working in a discipline like engineering where positivist worldviews of progress and forward growth in the name of Western science continually dominate curricula.

What would it look like, for instance, to incorporate relational approaches to undergraduate curriculum, where 'problem-solving' may shift into 'listening to and understanding' community needs and working with them to discover possible interventions? As Liboiron and Cotter pointed out,<sup>[66]</sup> how can researchers 'participate' with Indigenous groups on various methodologies and frameworks or even co-authorship and collaborators? The RESEAU Centre for Mobilizing Innovation<sup>[67]</sup> is an example of this kind of engagement with Indigenous groups, where water engineering helped to address the issues of clean drinking water on Indigenous reserves. RESEAU implemented the Community Circle, a platform inviting community members and all stakeholders to express their needs and aspirations. Not only did this project address the technical challenges of clean drinking water in rural communities, but it also incorporated other disciplinary approaches using research-based theatre as a methodology to educate and interact with communities.<sup>[68]</sup>

It seems like an opportune moment to re-vision engineering education especially with the integration of AI tools and the backdrop of complex challenges we face. We can re-examine the content of what we teach through the lens of core technical competencies and pedagogy that may reflect how multiplicity of understanding is integrated. There is also an opportunity to learn to 'co-learn' that is beyond simply working effectively in teams. There is a unique opportunity to prepare our students to be open and collaborative. They should recognize the value of relationship building and

fostering partnerships not only across disciplines but also cultures, and in doing so, they should value different knowledge systems.

## 4 | REFLECTIONS

In this concluding reflection, we aim to present considerations for fellow educators in chemical engineering, as well as those involved in engineering and STEM education more broadly. Our approach involves synthesizing the dialogue as a form of a 'results' section, and through these reflections, we intend to propose areas where future dialogue can extend beyond the confines of this paper. In addition to the synthesis of the dialogue, we also at times pose inquiry questions for others to contemplate. This section provides a comprehensive framing, while also highlighting synergistic ideas, concepts, concerns, and questions from leaders in the field.

### 4.1 | Impact of AI through design

We wonder how AI will change the pedagogy of chemical engineering education, especially in design. Would there be a limit in the need for procedural knowledge? The number of established processes that students design from scratch may become more limited, but as engineers they will still be involved in designing, design modification, and operation of processes and/or equipment. Students still need to be able to validate results obtained through a process simulator or other increasingly complex AI systems. In comparison, it will be less important for them to learn a 14-step process to design a shell and tube heat exchanger as a learning outcome in and of itself, as this may be easily automated. Allowing students to design it from scratch, however, gives them the confidence of actually designing something, and provides opportunities for them to build quantitative intuition.

As an educator who has taught AI in undergraduate and graduate courses for over 35 years, Venkatasubramanian explained how integrating AI modelling with first principle-based models is imperative in teaching AI in chemical engineering.<sup>[69]</sup> The mechanistic understanding based on first principles in physics, chemistry, and/or biology of our systems must govern any models that lead to hybrid AI models. Models based on symbolic AI (i.e., the methods of knowledge-based expert systems) can deal with complex and nonlinear systems, often seen in industrial processes, with incomplete and/or uncertain data. Coupled with data-driven numeric AI (i.e., machine learning), these hybrid AI models have been applied to materials science and design (e.g., polymers). Nevertheless, the importance of underlying physics and chemistry

in these design scenarios cannot be overstated. As AI models increasingly become virtual 'black boxes', it will become increasingly important for students to be able to use their knowledge to evaluate their outputs and guide them back to physical reality.

As educators, it would be beneficial to have clarity regarding the outcomes we are seeking to achieve. The desired outcome should not be students knowing the steps to design a heat exchanger; rather, it is building confidence in their capacity to transfer knowledge from theory to practice or to evaluate the reality and suitability of a design produced by an AI. To put it more generally, instructing procedural knowledge can still have a role as a mechanism for learning, but it is also important to recognize that the goal will no longer be for students to learn how to do a procedure themselves.

## 4.2 | Systems approaches

Through the instruction of thermodynamics, chemical engineers are introduced in their junior years to the reductionist approach of separating a system of interest from the surroundings. This approach gets reinforced through the creation of control volumes used to perform mass and energy balances. At the same time, chemical engineers are further equipped to think in systems as we are trained to zoom in and out of process flow diagrams, and to think across various temporal or spatial scales in a system so as to identify characteristic times or distances.

Thus, analyzing systems has long been part of learning chemical engineering but the reductionist mentality traditionally inherent to this instruction is increasingly less suited, as challenges become more complex. Bakshi asserted that a tension exists between reductionism of chemical engineering science and the holism of process systems engineering, and challenges that greater efficiency of a system may not be the only goal when considering, for example, planetary boundaries and circular economy.<sup>[70]</sup> Systems thinking and theory provides a way to understand, manage, and respond to complexity.<sup>[71]</sup>

Students would significantly benefit from learning broader systems perspectives (i.e., systems literacy, theory, and practice), and this type of instruction can be more than simply procedural (e.g., how to perform a life cycle analysis). It would expand the worldviews of students and ameliorate many of the challenges presented in the Dialogue. In doing so, it can potentially achieve four complementary learning orientations:

1. At an ethical level, learning systems perspectives can support sociotechnical analysis of the broader benefits

and impacts that may arise from creating or implementing a technology. As earlier noted, technical problems are situated within environmental, social, economic and political contexts, and it is important for our students to understand the role of engineers in navigating these. When asking students to conduct a techno-economic analysis, how do we guide them to situate it in the context of a wider system? How should the net social benefit be considered and not simply fall off the flow sheet?

2. At an epistemological level, applying a broader systems approach can help students learn about and become comfortable with using different types of non-technical knowledge. While engineers work in a predominantly technical domain, the successful design and sustainable implementation of technical solutions requires collaboration with experts in other domains. Our students need to recognize that the 'engineers rule the world' trope is antiquated. How might we better teach them how inter- and trans-disciplinary approaches can present expansive ways of identifying and offering technological opportunities and challenge-solutions?
3. At a metacognitive level, using a broader systems approach can help students learn knowledge transfer by encouraging them to make connections across different clusters of knowledge. How can we more effectively incorporate this meta level learning outcome into our instruction? Certainly, through systems mapping such as concept maps and causal loop diagrams, we can help them understand and identify the interrelations between elements/concepts and feedback behaviours. What other tools and instructional strategies can we bring in to help them make connections across systems of systems and learn to intentionally mobilize their knowledge well beyond the context through which it was originally introduced?
4. At an ontological level, using systems perspectives can support grappling with complexity and uncertainty, as well as understanding and embodying multiple worldviews or multiple worlds.<sup>[72]</sup> This approach encourages active engagement with and intervention in the world, encapsulated by the term *systems beings*,<sup>[73]</sup> which signifies the multifaceted experience and awareness of existing within and learning about systems. Systems beings, in essence, can be regarded as an ontology that recognizes our inherent interconnected nature in and with systems.

The concept of Industry 4.0 was introduced a decade ago to describe the increased integration of cyber-physical systems into industrial production and automation.<sup>[74,75]</sup> This has now been superseded by 'Industry 5.0 or Society

5.0' to describe solving socio-technical challenges and opportunities through the integration of physical and virtual spaces. The vision of Industry 5.0 is 'an innovative, resilient, socio-centred and competitive industry, which respects planetary boundaries and minimizes its negative environmental impact' (p. 3).<sup>[76]</sup> Systems perspectives will be central to bridging between these physical and virtual spaces and addressing the socio-technical dimensions and gaps. As earlier discussed, transdisciplinary instruction can promote this type of broader more holistic approach: 'The products of the fifth industrial revolution, cyber-physical systems, cannot be designed and developed without transdisciplinary environments' (p. 5).<sup>[56]</sup>

### 4.3 | Productive tensions between technical knowledge and other ways of knowing

Through our consideration of Question 2 (Section 3.2), we noted a false paradox between helping students to value and work with multiple types of knowledge and the need for them to identify what is more credible or appropriate evidence in a given context. Traditional engineering education can instil an embedded bias to privilege technical knowledge and quantitative evidence. This is understandable since engineering has traditionally focused more on the inanimate, where quantitative evidence is generally more appropriate to support claims, conclusions, and recommendations. However, analysis of animate dimensions has become an increasingly important part of implementing technologies requiring considering human relations, perspectives, and values.

As we previously noted, our students need to be able to recognize, assess, and integrate many types of evidence and recognize which is best suited for a given context. As a starting point they will need to move beyond a binary or dualist perspective, to what Perry referred to as 'commitment within relativism'.<sup>[77]</sup> In addition they will need to be able to critically assess the available evidence in relation to the knowledge paradigm from which it originates. For example, valuable qualitative evidence can emerge from social science surveys of the perceived risk, benefits, or impacts of a new technology (e.g., windmills), the appearance of a new product being introduced to the marketplace (e.g., a smartphone), the acceptance of a health intervention (e.g., a vaccine), or the desirability of a proposed behaviour change (e.g., cycling or public transit instead of driving a car). From this, students will be able to recognize that not all evidence is equally valid or carries equal weight in a given context, and that non-technical dimensions may at times have a larger influence on the success of a technology.

Graduates may be increasingly called upon to communicate their recommendations in accessible and persuasive ways in order to counter increased misinformation. For example, while western medicine has made incredible achievements in terms of disease treatment, other approaches may offer options in terms of disease prevention. Moreover, what may be true on average across a population is not necessarily applicable to any specific individual. Thus, evaluating this type of health-care evidence is challenging, encouraging misguided relativist views that all evidence is of equal value. This uncertainty left the door open to types of conspiracy theory-based pseudo-science such as vaccine opposition and climate change denial.

While the challenge of misinformation and 'alternate facts' goes well beyond the domain of chemical engineers alone, as future stewards of technology we have a responsibility to prepare students for the role that they will play in society. One approach may be to cultivate reflexivity in ourselves and our students by asking the epistemic question: 'how do we know what we know?' and asking the ontological question: 'how does our shifting knowledge integrate with different worldviews?' We can create a shared experience of exploring our worldviews and become more aware of the engineering knowledge in the context of others. This can also include active engagement of understanding in diverse or Indigenous knowledge systems. As instructors, we can exercise our reflexivity in terms of how we are teaching and at times modelling this to our students. Guiding reflexive dialogue in our classrooms may also help students broaden their awareness of different types of evidence and knowledge. We can encourage and model critical analysis of evidence, including that generated by AI. By creating a culture of reflexive learners, we allow us to be aware and embrace diversity, and be more adaptive in facing complex challenges.

Creating conditions for engineers to apply different methods, theories, and tools that can be applicable to their education as well as their professional future remains essential in current education programs. As noted earlier, Indigenization of the engineering education programs in Canada has been mandated by Engineers Canada.<sup>[78]</sup> For many, this presents both an exciting and challenging proposition. It fundamentally requires adopting a variety of different epistemologies (ways of knowing and learning about knowledge), ontologies (ways of being or understanding reality), and methodologies (ways of doing the practice). This will not happen overnight and requires systematic training for educators to incorporate curricula into existing courses as well as motivating programs to create new courses.



#### 4.4 | Supporting student comfort with uncertainty

Increased complexity inevitably increases uncertainty. How is engineering education adapting to help students deal with the greater uncertainty, which includes holding space for paradoxes and contradictions? We have all experienced living in uncertainty and complexity through the pandemic, and continue to learn from the impact. As Talanquer et al.<sup>[79]</sup> have documented for chemistry education, might there be an opportunity for chemical engineering education to reconsider/reimagine the curriculum in light of facing this volatility, uncertainty, complexity, and ambiguity (VUCA)?

From the perspective of placing students outside of their comfort zone for learning, Taylor and Baker<sup>[80]</sup> described creating dissonance—that is, encountering inconsistencies between two or more thoughts, beliefs, or experiences, followed by a meaning-making process—that can support learning and development. There may be a way to practice holding space for multiple truths or building capacity to be in cognitive dissonance in engineering education in order to cultivate a certain type of stamina to face uncertainties.

In addition to increasing stamina and comfort with uncertainty, how can we help our students recommend courses of action, or navigate, despite inevitable uncertainty? On one hand, we need to build self-confidence as doubt from uncertainty can paralyze action resulting in lost opportunities for progress. On the other hand, inaction may in fact be the wise choice in some cases, when the risks outweigh the potential benefits, or when these risks and benefits are distributed inequitably. Thus, navigating uncertainty is closely related to managing risk. Secondly, navigating uncertainty can be value-laden, creating biases towards more familiar knowledge—technical knowledge, in the case of engineers. Additional cognitive biases can be introduced by a sense of identity and community affiliation, such as their identity as an engineer.<sup>[81]</sup>

Being comfortable in a highly uncertain environment is essential to an entrepreneurial mindset. Helping students understand and manage uncertainty can also yield gains in terms supporting entrepreneurship. In a collaborative learning setting, as many engineering students are often engaged in, there are opportunities to experience relational and epistemological uncertainties.<sup>[82]</sup> Relational uncertainties occur around social interactions, while epistemological uncertainties invoke being unsure about the content are present in the process of collaborative work. These uncertainties can be a motivation for learning.

#### 4.5 | International considerations

Chemical engineering will continue to have an essential role in addressing and leveraging modern challenges and opportunities. These roles may differ across the globe. While Canada has entered a post-industrial era and economy, countries in the Global South are still growing their industrial base as part of a desire to expand economies and increase equitable opportunities for citizens.

Many Canadian engineering schools attract international students who decide to join based on the prestige of North American institutions. It is not uncommon for these students to arrive with expectations shaped by the industrial and economic aspirations of their home countries, and many may be anticipating a curriculum with a traditional manufacturing focus, especially around the oil and gas industry. While we forge ahead on the path to decarbonization, do we risk neglecting the legacy industries of chemical engineering that continue to grow in other parts of the world? For example, while Canada has placed a high value on rapid decarbonization with a transition to new energy sources and related jobs, and this is expected to occur within the career lifetime of current students, this is not necessarily the case in some other countries that are still seeking to use fossil fuels to build their economies and where related jobs are expected to be available for many decades. What might this mean for future curricula?

As we reflect on the proposed evolution of chemical engineering education, this raises a compelling question around the contrasting values of an institution and its diverse student body. Perhaps Canada should develop a wider range of chemical engineering curricula across its schools or streams within programs, so that students seeking more traditional or more modern learning can find places that meet their needs and interests.

### 5 | CONCLUSION

Operating both as a theoretical framework and a practical discipline for professionals and researchers, chemical engineering must adapt to changing complex systems. The field requires higher level cognitive and metacognitive skills to navigate the intricacies of data, AI, machine learning, as well as the pressing societal and human needs related to energy, commodities, connectivity, and self-actualization. Moreover, there is an urgent call for relational accountability concerning nature and the environment.

Navigating this multifaceted landscape requires not only collaborative but also flexible transdisciplinary engagement. The challenges at hand, such as the imperative to

limit global temperature rise to 1.5°C, are further complicated by the reluctance to alter consumption patterns or restrict economic growth. Recognizing that an elegant technical solution alone is insufficient due to continuously changing boundary conditions and the inability of models to encompass all relevant factors, it becomes evident that the predicament stems from centuries of evolution within socio-economic and political systems driven by technological innovation. Consequently, the resolution of these challenges demands more than just technological innovation and advancement.

There is a unique opportunity to reshape chemical engineering education to address the increasingly complex world. The boundaries between engineering disciplines, or even engineering, physical, medical, and social sciences, are becoming more diffuse, blurring the traditional meanings of these disciplinary distinctions. Moreover, most opportunities are emerging at the interfaces between the traditional disciplines, effectively driving a convergence. While the traditional identity of a chemical engineer may be dissolving away into this increasingly complex knowledge mixture, knowledge and perspectives related to chemical engineering remain essential to technological and societal advancement.

While the future remains unpredictable, especially in an era of rapid change, we have the capacity to prepare students in a manner that equips them for a multitude of potential futures. The objective is not to pinpoint a singular, most probable path, but rather to acknowledge the existence of numerous possible trajectories and identify common educational elements that can be widely applied. Revisiting the concept of ‘futuring’ introduced at the outset of this paper, and practiced throughout through a dialogue as both pedagogy and educational research, we recognize that we all possess the ability to collaboratively envision alternative worlds. Given that engineering inherently involves imagining and designing futures, we are well-positioned to actively engage in these generative dialogues.

## AUTHOR CONTRIBUTIONS

**Pranav Chintalapati:** Writing – review and editing; investigation; writing – original draft. **Greg Evans:** Writing – review and editing; literature review; investigation; writing – original draft. **Derek Gladwin:** Investigation; conceptualization; writing – review and editing; literature review; methodology. **Marnie Jamieson:** Investigation; writing – original draft; writing – review and editing. **Naoko Ellis:** Conceptualization; investigation; writing – original draft; writing – review and editing; literature review; project administration.

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Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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